EXERCISE 5

Saddle Point Invariant Manifold and Asymptotic Decay

Original statement · ProblemsCh3.pdf, p. 2

Phase Plane • Stable Separatrix

PHASE 1: STATEMENT & SETUP

If the roots of the auxiliary equation are $k_1 > 0$ and $-k_2 < 0$ then the solution is $x(t) = Ae^{k_1 t} + Be^{-k_2 t}$. For most choices of initial conditions $x(0) = x_0,\ \dot{x}(0) = y_0$, we will have $x(t) \to \pm\infty$ as $t \to \infty$. However, there are some special initial conditions for which $x(t) \to 0$ as $t \to \infty$. Find the relationship between x_0 and y_0 that ensures this.

PHASE 2: FUNDAMENTAL FORMULATION

Given $k_1 > 0$ and $k_2 > 0$, the mode $e^{k_1 t}$ diverges exponentially to $+\infty$, while $e^{-k_2 t} \to 0$ as $t \to \infty$. Therefore, $x(t) \to 0$ if and only if the unstable amplitude is strictly annihilated: $A = 0$.

METHOD OF THE CLASS NOTES [NOTES, P. 5]

Case 01 of the notes with k_1 and $-k_2$, and the initial data matched with the matrix of the notes:

$$\begin{pmatrix} 1 & 1 \\ k_1 & -k_2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \qquad W(0) = -k_2 - k_1 \neq 0.$$

Cramer's rule gives $A = \frac{-k_2 x_0 - y_0}{-k_1 - k_2} = \frac{k_2 x_0 + y_0}{k_1 + k_2}$, so $x(t) \to 0$ exactly when $A = 0$, that is $y_0 = -k_2 x_0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Step 1: Differentiate the general solution:

$$x(t) = Ae^{k_1 t} + Be^{-k_2 t} \implies \dot{x}(t) = k_1 Ae^{k_1 t} - k_2 Be^{-k_2 t}$$

Step 2: Impose initial Cauchy conditions at $t = 0$:

$$x(0) = A + B = x_0 \tag{1}$$

$$\dot{x}(0) = k_1 A - k_2 B = y_0 \tag{2}$$

Step 3: Solve for the amplitude A:

Multiply (1) by k_2 and add to (2):

$$k_2(A + B) + (k_1 A - k_2 B) = k_2 x_0 + y_0 \implies (k_1 + k_2)A = y_0 + k_2 x_0$$

$$A = \frac{y_0 + k_2 x_0}{k_1 + k_2} \tag{3}$$

Step 4: Asymptotic Condition $x(t) \to 0$:

Because $\lim_{t\to\infty} e^{k_1 t} = \infty$ and $\lim_{t\to\infty} e^{-k_2 t} = 0$:

$$\lim_{t\to\infty} x(t) = 0 \iff A = 0$$

$$\frac{y_0 + k_2 x_0}{k_1 + k_2} = 0 \implies \mathbf{y_0 = -k_2 x_0}$$

PHASE 4: INTERPRETATION & DYNAMICAL SYSTEMS

ASYMPTOTIC DECAY CONDITION

$$\boxed{y_0 = -k_2 x_0 \iff \dot{x}(0) = -k_2 x(0)}$$

When this relationship holds, $A = 0$ and $B = x_0$, giving the purely decaying trajectory $\mathbf{x(t) = x_0 e^{-k_2 t}}$.

GEOMETRIC PHASE PLANE MEANING: SADDLE SEPARATRIX

In the phase plane $(x, \dot{x})$, the origin is a saddle equilibrium. The line $\dot{x} = -k_2 x$ is the **stable invariant manifold** (separatrix). Any state starting off this 1D manifold is swept away by the unstable manifold $\dot{x} = k_1 x$, diverging exponentially to infinity.